

1. Lee, an employee of the company, wants to buy a Mac laptop. This matter needs the approval of leaders. Each leader has different authorities, and the maximum amount that can be approved is limited. Lee first applied to the group leader, but because the amount was too large, it exceeded the approval authority of the group leader. So the group leader went to his leader, the Department Manager. But the Department Manager still doesn't have enough power. So the Department Manager found the CFO, who had enough power to pass the application. Finally, Lee received a grant to buy a Mac laptop. (note: The amount approved by the group leader is within 1000, the amount approved by the Department Manager is within 10000, the amount approved by the CFO is within 50000, and the price of the computer purchased by Lee is 11000)

BudgetHandler

```
package experiment22_1;

interface BudgetHandler {

    boolean handle(int amount);

    void setNextHandler(BudgetHandler budgetHandler);

}
```

CFO

```
package experiment22_1;

class CFO implements BudgetHandler{

    private BudgetHandler nextHandler;

    @Override

    public void setNextHandler(BudgetHandler nextHandler)

    {

        this.nextHandler = nextHandler;

    }

}
```

```

@Override

public boolean handle(int amount) {

    if(amount <= 50000){

        System.out.println("CFO approved the budget for:
" + amount);

        return true;

    }else{

        if(nextHandler != null){

            return nextHandler.handle(amount);

        }

    }

    return false;

}

}

```

Experiment22_1

```

package experiment22_1;

public class Experiment22_1 {

    public static void main(String[] args) {

        GroupLeader leader = new GroupLeader();

        Manager manager = new Manager();
    }
}

```

```

CFO cfo = new CFO();

leader.setNextHandler(manager);
manager.setNextHandler(cfo);

System.out.println(String.format("I needs to buy a
Mac laptop with a budget of: %d", 11000));

if (leader.handle(11000)) {
    System.out.println("thanks");
} else {
    System.out.println("unfortunately");
}
}
}

```

GroupLeader

```

package experiment22_1;

class GroupLeader implements BudgetHandler{
    private BudgetHandler nextHandler;

    @Override
    public void setNextHandler(BudgetHandler nextHandler)

```

```

{
    this.nextHandler = nextHandler;
}

@Override
public boolean handle(int amount) {
    if(amount <= 1000){
        System.out.println("GroupLeader approved the
budget for: " + amount);
        return true;
    }else{
        if(nextHandler != null){
            return nextHandler.handle(amount);
        }
    }
    return false;
}
}

```

Manager

```

package experiment22_1;

class Manager implements BudgetHandler{

```

```
private BudgetHandler nextHandler;

@Override

public void setNextHandler(BudgetHandler nextHandler)
{

    this.nextHandler = nextHandler;

}

@Override

public boolean handle(int amount) {

    if(amount <= 10000){

        System.out.println("Manager approved the
budget for: " + amount);

        return true;

    }else{

        if(nextHandler != null){

            return nextHandler.handle(amount);

        }

    }

    return false;

}

}
```

Output

```
D:\JDK\bin\java.exe "-javaagent:D:\idea\IntelliJ IDEA
I needs to buy a Mac laptop with a budget of: 11000
CFO approved the budget for: 11000
thanks

Process finished with exit code 0
```